

COMPLEXITY AND INFORMATION-BASED ANALYSIS OF THE HEART RATE VARIABILITY (HRV) WHILE SITTING, HAND BIKING, WALKING, AND RUNNING

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Abstract

Analysis of the variations of heart activity during different human activities is an important area of research in sport sciences. Therefore, in this paper, we evaluated the variations of heart activity for 23 subjects while sitting, hand biking, walking, and running. Since the obtained R-R time series (as the indicator of heart rate variability (HRV)) has a complex structure that contains information, we employed fractal dimension, sample entropy, and Shannon entropy for our analysis. According to the results, doing a harder activity causes a more significant

alteration in the complexity and information content of HRV. The results of statistical analyses also verified the obtained results. Similar investigations can be conducted in case of other activities to evaluate the variations in heart activity in different conditions.

Keywords: Heart Rate Variability (HRV); Complexity; Information; Fractal Dimension; Sample Entropy; Shannon Entropy.

1. INTRODUCTION

Analysis of the variations of physiological conditions of humans during different activities is an important area of research under sport sciences. Since the heart highly reacts to human movements, therefore, many researchers have focused on the analysis of heart rate variability (HRV) in different sports activities. For instance, we can call the studies that analyzed HRV amongst cyclists under perceived variations of risk exposure,¹ investigated HRV in cross country cycling athletes,² studied correlation properties of HRV during endurance exercise,³ evaluated the association between walking capacity and HRV in patients with symptomatic peripheral artery disease,⁴ investigated the relation between HRV and training load in interval running exercises,⁵ evaluated the changes in HRV with respect to exercise intensity and time during treadmill running,⁶ investigated the effects of passive simulated jogging on short-term HRV,⁷ and analyzed HRV of healthy people and cardiac patients at rest and during aerobic exercise.⁸

It is known that HRV in the form of the R-R time series has a complex structure.⁹ Therefore, fractal theory is a proper technique to analyze its complexity. In general, fractal objects have repeating patterns that are distributed on every scale inside them. The complexity of these objects is quantified using the fractal dimension. For years, many researchers have applied fractal theory to analyze the complexity of different physiological and biological time series (e.g. EEG signals,¹⁰ PCG signals,¹¹ EMG signals,¹² GSR,¹³ MEG signals,¹⁴ DNA walk¹⁵) and images (e.g. human face,¹⁶ X-ray images¹⁷).

Specifically, some researchers investigated the complex structure of HRV signals using the fractal theory. The reported works that analyzed the HRV of normal subjects of different ages,¹⁸ predicted cardiac death,¹⁹ evaluated the effects of pharmacological adrenergic and vagal modulation on heart

rate dynamic,²⁰ analyzed the variations of heart rate during non-REM sleep,²¹ investigated the variations of heart rate for patients with the peripheral arterial disease,²² diabetes,²³ Chronic Obstructive Pulmonary Disease (COPD),²⁴ and evaluated the correlation between the complexities of HRV and walking paths²⁵ using fractal theory, are worthy of being mentioned.

Besides fractal analysis, other nonlinear analysis techniques such as approximate entropy and sample entropy also can be used to evaluate the complex structure of HRV signals. Sample entropy as a measure of complexity does not depend on the data length.²⁶ Since R-R time series of different subjects (extracted from ECG signals) have different lengths while doing different activities, the calculation of sample entropy helps us to verify the result of the fractal analysis, which is dependent on the length of data. Sample entropy has been applied widely for analyzing the complexity of various types of physiological and biological time series.^{27–31} Specifically, the applications of sample entropy in the analysis of HRV^{32–34} have been extensive.

Besides, R-R time series (as HRV signals) contains information,³⁵ and therefore, Shannon entropy can be employed to analyze its changes in different conditions. Similarly, Shannon entropy has been applied widely to analyze the information content of various types of physiological and biological time series.^{36–40} Specifically, the applications of Shannon entropy in the analysis of HRV^{41–43} have been extensive.

Therefore, in this research, we apply fractal analysis, sample entropy, and Shannon entropy to investigate the alterations in HRV in the case of different activities of humans.

In the following sections, firstly, we discuss our methods of analysis. Then, we present the database and steps of analysis. Finally, we bring the results and discuss them, which also include some future works.

2. METHOD

In this study, we want to investigate the variations in heart activity while sitting, hand biking, walking, and running. Therefore, we analyze ECG signals as the feature of heart activity. R-R time series (obtained by extracting R peaks of ECG signals) have complex nonlinear structures. Therefore, to analyze the alterations of HRV in various conditions, we run fractal analysis and compute the fractal dimension as the complexity index. In this study, we calculate the fractal dimension of the R-R time series using the box-counting algorithm.

This algorithm segments a signal using boxes that have the same size (μ). Then, it counts the number of these boxes (n). The algorithm changes the size of these boxes in several iterations and finally computes the fractal exponent (F) using the number and size of boxes in each step (Eq. (1)):

$$F = \lim_{\mu \rightarrow 0} \frac{\log n(\mu)}{\log 1/\mu}. \quad (1)$$

Equation (2) shows the general form of fractal dimension,⁴⁴ when t is the order of F , and p_i indicates the probability in the i th segment of the signal, that itself is calculated using Eq. (3).

$$F_t = \lim_{\mu \rightarrow 0} \frac{1}{t-1} \frac{\log \sum_{i=1}^n p_i^t}{\log \mu}, \quad (2)$$

$$p_i = \lim_{y \rightarrow \infty} \frac{x_i}{y}, \quad (3)$$

x_i is the number of occurrences in the i th segment along the whole length of the signal (y).

On the other hand, considering the same period of each activity for all subjects, the extracted R-R time series from their ECG signals have different lengths. Therefore, to overcome its effect on the calculations, we computed the sample entropy of the R-R time series in various conditions. Like the fractal dimension, a bigger value of sample entropy indicates higher complexity.⁴⁵ Therefore, sample entropy is employed to verify the results from the fractal analysis.

Considering a signal in the form of $\{r(1), r(2), r(3), \dots, r(n)\}$ with a constant interval of α , we define a template vector of length z (embedding dimension) in the form of $R_z(i) = \{r_i, r_{i+1}, r_{i+2}, \dots, r_{i+z-1}\}$, and the distance function $d[R_z(i), R_z(j)]$ ($i \neq j$) is to be Chebyshev distance. Then, the sample entropy (SamEn) is formulated

as⁴⁵:

$$\text{SamEn} = -\log \frac{B}{C}. \quad (4)$$

Considering ε as the tolerance ($0.2 \times$ standard deviation of data), B stands for the number of template vector pairs that:

$$d[R_{z+1}(i), R_{z+1}(j)] < \varepsilon. \quad (5)$$

Besides, C stands for the number of template vector pairs that:

$$d[R_z(i), R_z(j)] < \varepsilon. \quad (6)$$

Besides, since R-R time series contain information, we also used Shannon entropy to investigate the alterations of the information content of HRV between different activities. Shannon entropy is the indicator of average embedded information in a message. In fact, a message in this research is considered an R-R time series. A larger Shannon entropy demonstrates larger information in a message.⁴⁶ Shannon entropy is formulated as⁴⁷:

$$H = - \sum_i p_i \log p_i, \quad (7)$$

where in this research, p_i indicates the probability of HRV with the value of a_i , where i can be changed along with the signal.

Therefore, we calculated the fractal dimension, sample entropy, and Shannon entropy to evaluate the alterations of the complexity and information content of HRV in different activities.

3. DATABASE AND ANALYSIS

In this research, we used the open-access database provided in Ref. 48. This database contains ECG signals of 25 subjects who performed five different tasks (sitting, a maths test on a tablet, walking on a treadmill, running on a treadmill, using a handbike) for 2 min. Exercise chest strap ECG which resembles approximately V2-V1 with the ECG amplifier directly mounted on the strap, which allowed the detection of R peaks at very high precision even while running. The data has been recorded at the sampling frequency of 250 Hz. Since we are interested in analyzing HRVs while doing different sport activities, we did not analyze the HRVs of subjects while doing a math test. Figure 1 shows different scenes belong to the recorded videos from a subject during different tasks.



Fig. 1 Different scenes of data collection during sitting (a), hand biking (b), walking (c), and running (d).⁴⁸

Since this database includes the extracted R peaks from ECG signals of subjects in different activities, we designed an algorithm that generated the R-R time series by calculating the time difference between every two consecutive R peaks. A sample raw ECG signal during running and generated R-R time series are shown in Fig. 2. It should be noted that the number of R peaks in the case of different subjects was different.

We calculated the fractal dimension and sample entropy for the generated R-R time series. As was mentioned previously, the box-counting algorithm was chosen for the fractal analysis. It should be noted that all these analyses were conducted in MATLAB (MathWorks, USA). Here, we should note that we excluded the recorded data for two subjects from our analysis since the extracted R

peaks were not provided for these subjects during two sessions.

We analyzed the significance of heart rate alterations between various conditions by running the one-way repeated measures ANOVA test ($\alpha = 0.05$). We also ran posthoc Tukey tests and effect size analyses ($\alpha = 0.05$) for pairwise comparisons between different conditions.

4. RESULTS

Figure 3 shows the fractal dimension of the R-R time series during sitting, hand biking, walking, and running. Error bars indicate standard deviations.

According to the result of the ANOVA test (P -value = 0.0001 and F -value = 36.7286), the variations of the fractal dimension of the R-R time

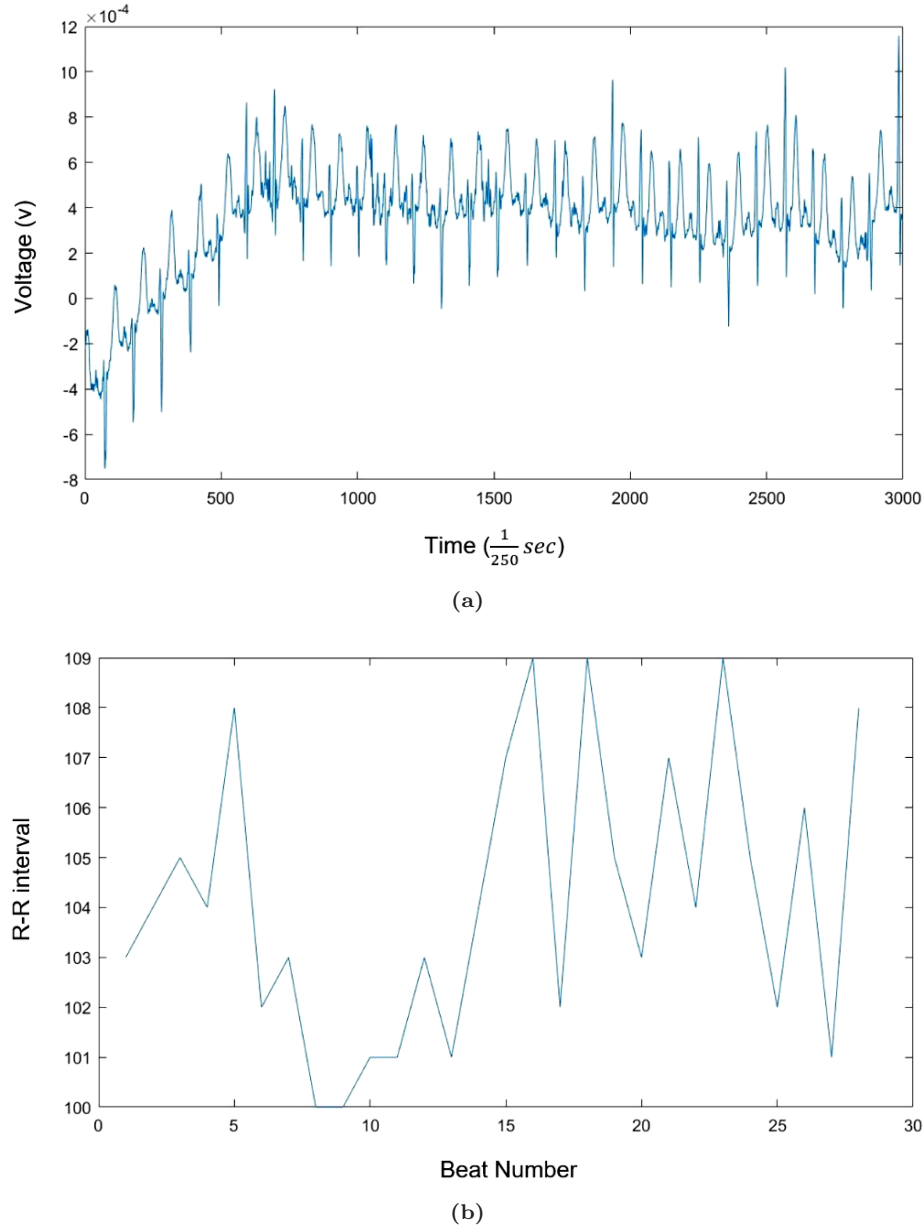


Fig. 2 Sample of raw ECG signal during running (a) and the generated R-R time series (b).

series between different activities were significant. As Fig. 3 shows, the R-R time series of subjects had the greatest fractal dimension values during sitting (rest). On the other hand, by moving to hand biking, walking, and running in Fig. 3, the fractal dimension values reduce. Since the fractal dimension indicates the complexity of the R-R time series in this research, therefore, we can state that during rest (sitting), the R-R time series has the highest complexity, which respectively reduces in hand biking, walking, and running. Therefore, based on the results, working on a harder activity causes a greater alteration in the complexity of HRV.

The results of the posthoc test in Table 1 indicate that the alterations of the complexity of the R-R time series between sitting versus running, hand biking versus running, and walking versus running were significant. According to the results, a harder activity (running, walking, and hand biking, respectively) causes a more significant alteration in the complexity of HRV. The results of the effect size analysis in this table indicate that running, walking, and hand biking respectively had the greatest to lowest effect on altering the complexity of HRV.

As was mentioned before, since fractal dimension depends on the length of data, and different subjects

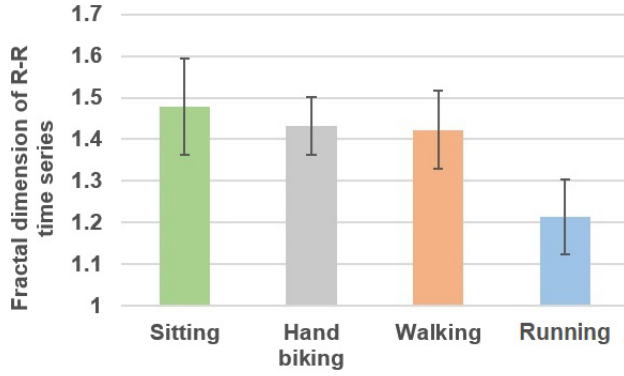


Fig. 3 The fractal dimension of the R-R time series in different conditions.

Table 1 Comparison of the Fractal Dimension of R-R Time Series Between Different Conditions.

Pair	<i>P</i> -Value	Cohen's <i>d</i>
Sitting versus Hand biking	0.3368	0.5075
Sitting versus Walking	0.1770	0.7025
Sitting versus Running	0.0001	2.5790
Hand biking versus Walking	0.9832	0.1214
Hand biking versus Running	0.0001	2.0880
Walking versus Running	0.0001	2.1912

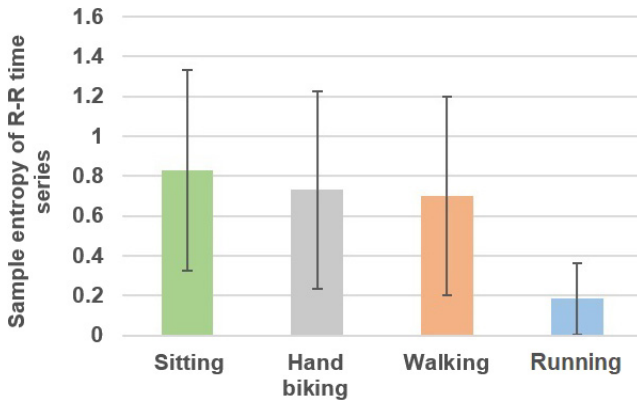


Fig. 4 The sample entropy of the R-R time series in different conditions.

had different lengths of R-R time series, we also calculated the sample entropy of R-R time series to verify the obtained results from the fractal analysis. Figure 4 shows the sample entropy of the R-R time series during sitting, hand biking, walking, and running.

Based on the result of the ANOVA test (P -value = 0.0001 and F -value = 9.9143), the variations of sample entropy of R-R time series were significant between different activities. As can be seen in Fig. 4,

during sitting, the R-R time series of subjects had the greatest values of sample entropy. On the other hand, by moving to hand biking, walking, and running in Fig. 4, the values of sample entropy reduce. Since sample entropy indicates the complexity of the R-R time series, therefore, we can state that during rest (sitting), the R-R time series has the highest complexity, which respectively reduces in hand biking, walking, and running. Therefore, based on the results, working on a harder activity causes a greater alteration in the complexity of HRV.

The obtained results from the posthoc test in Table 2 indicate the significant difference in the complexity of the R-R time series between sitting versus running, hand biking versus running, and walking versus running. Based on the results, working on a harder activity (running, walking, and hand biking, respectively) causes a more significant alteration in the complexity of HRV. Besides, the effect sizes in this table indicate that running had a greater effect on the variations of the complexity of HRV than walking, which itself had a greater effect than hand biking. Besides, as it was expected, sitting had the least effect on the alterations of the complexity of HRV. Therefore, the results of statistical analysis verify the obtained results in Fig. 4.

Comparing Fig. 4 with Fig. 3 indicates that both figures follow the same trend. Therefore, the results of the analysis of sample entropy verify the obtained results from the fractal analysis.

As was mentioned previously, we also calculated the Shannon entropy of the R-R time series in different conditions. The results of the analysis are provided in Fig. 5.

Based on the result of the ANOVA test (P -value = 0.0001 and F -value = 27.6947), the alterations of the Shannon entropy of the R-R time series between different activities were significant. As Fig. 5 demonstrates, the R-R time series of subjects

Table 2 Comparison of the Sample Entropy of R-R Time Series Between Different Conditions.

Pair	<i>P</i> -Value	Cohen's <i>d</i>
Sitting versus Hand biking	0.8712	0.1984
Sitting versus Walking	0.7603	0.2549
Sitting versus Running	0.0001	1.7059
Hand biking versus Walking	0.9962	0.0574
Hand biking versus Running	0.0004	1.4659
Walking versus Running	0.0008	1.3834

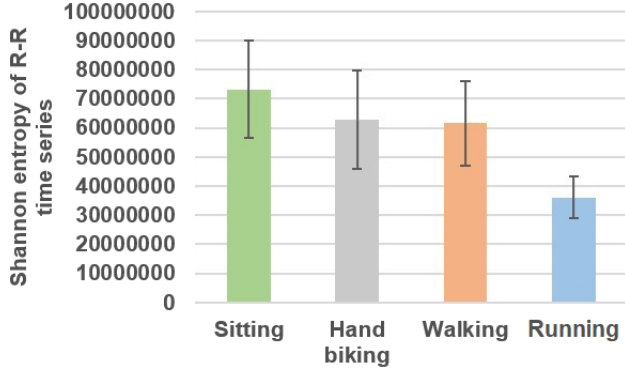


Fig. 5 The Shannon entropy of the R-R time series in different conditions.

had the greatest values of Shannon entropy while sitting. According to this figure, by shifting toward hand biking, walking, and running, the values of Shannon entropy reduce. Since Shannon entropy indicates the information content of the R-R time series in this research, therefore, we can state that during rest (sitting), the R-R time series contains the largest information, which respectively reduces in hand biking, walking, and running. Therefore, based on the results, working on a harder activity causes a greater alteration in the information content of HRV.

The results of the posthoc test in Table 3 indicate the significant difference in the information content of the R-R time series between sitting versus walking, sitting versus running, hand biking versus running, and walking versus running. As can be seen in this table, working on a harder activity (running, walking, and hand biking, respectively) causes a more significant alteration in the information content of HRV. Besides, the effect sizes in this table indicate that running had a greater impact on the alterations of the information content of HRV than walking, which itself had a greater effect than hand biking. Therefore, the results of statistical analysis verify the obtained results in Fig. 5. Besides,

Table 3 Comparison of the Shannon Entropy of R-R Time Series Between Different Conditions.

Pair	P-Value	Cohen's <i>d</i>
Sitting versus Hand biking	0.0810	0.6110
Sitting versus Walking	0.0361	0.7450
Sitting versus Running	0.0001	2.8897
Hand biking versus Walking	0.9877	0.0882
Hand biking versus Running	0.0001	2.0638
Walking versus Running	0.0001	2.2165

comparing Fig. 5 with Figs. 3 and 4 indicates that all figures follow the same trend.

Therefore, based on the obtained results, fractal dimension, sample entropy, and Shannon entropy are proper techniques that can decode the alterations of HRV between different activities.

5. DISCUSSION AND CONCLUSION

In this paper, we analyzed the alterations of HRV between different human activities. For this purpose, we ran complexity and information-based analyses. We computed the fractal dimension, sample entropy, and Shannon entropy of the R-R time series while subjects were sitting, hand biking, walking, and running.

According to the results, the R-R time series of subjects had the greatest values of fractal dimension, sample entropy, and Shannon entropy while sitting (rest). On the other hand, during hand biking, walking, and running, the R-R time series respectively obtained lower values of fractal dimension, sample entropy, and Shannon entropy. In other words, performing a harder activity caused a greater change in the complexity and information content of HRVs. We also compared the alterations of the fractal dimension, sample entropy, and Shannon entropy of the R-R time series between different activities using the posthoc test. Based on the results, running caused significant changes in the complexity and information content of the R-R time series compared to other activities. Besides, the results of effect size analysis indicated that running (as the hardest activity compared to others) had the most significant effect on alterations of the HRV.

Utilizing sample entropy to validate the fractal analysis results specifically was used here since the data length was short and different in the case of different subjects. Here, we should note that although fractal analysis is a robust technique and widely used to analyze the complexity of HRV, however, due to its sensitivity to the length of data, it is advised that its results been verified with other techniques that are not sensitive to the length of data (e.g. sample entropy). This consideration does not mean that fractal theory is not proper for complexity-based analysis. As was mentioned before, the fractal theory has been widely used to analyze different types of time series, specifically when comparisons were made between different

conditions that resulted in the same and long length of data.

The conducted analysis in this research could be extended in case of other human activities to investigate how heart activity changes in case of different activities. Besides, we can run similar analyses in the case of patients with different heart diseases (e.g. coronary artery disease,⁴⁹ arrhythmia⁵⁰) to investigate how the HRV of patients changes while doing different activities. We can also analyze the effect of intensity of each activity on the HRV of subjects. For instance, we can run a similar analysis when subjects perform hand biking, walking, and running at different speeds and accordingly evaluate how the changes in the intensity of activity affect the variations of heart activity.

It should be noted that these analyses are not limited to study the HRVs of subjects. For instance, we can expand these analyses to evaluate the changes in muscle and brain reactions by analysis of EMG and EEG signals to investigate how the muscle and brain react when the activity becomes harder. Besides, as an interesting area of research, we can analyze the correlation between the activities of different organs while doing these activities. For instance, we can simultaneously record and analyze EEG and EMG signals of subjects while doing different activities and investigate how strongly is the correlation between brain and muscle reactions. Since the brain controls the activities of different organs, a correlation should exist.

Overall, all these analyses have great importance in physiological and sport sciences and will help us to improve performance during different sports.

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